SYNAPSE | LEARNER'S WORLD

The Learner's World Plan

The problems worth solving, everything to build, a fight with the plan, and the path to $3B ARR

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Companion: Synapse Data Collection Master Register

1. The Verdict (read this first)

Master Plan v2 describes a Learner World with ten spaces and roughly sixty features. As a map of what is possible, it is excellent. As a plan for what to build first, it will fail: it is a Notion, NotebookLM, Gauth, Duolingo and ChatEdu rolled into one, built by a small team, for learners on cheap phones with little data. This document keeps the whole vision, ranks every piece, attacks it, and shows the version that can actually work.

The five conclusions

- **The expensive problem is not "no app".** It is high-stakes exams (Grade 7, 9, 12) that decide a life path, faced with too little feedback, too few materials, no money for tutors, a second language of instruction, and a phone with little data and unreliable power.
- **The wedge is "Pass your ECZ exam".** Past papers with worked solutions, honest diagnostics, spaced practice, local-language explanations, offline, paid in tiny mobile-money amounts. Exam prep is the one thing families already pay for (tuition, photocopied bundles, rewrites). Everything else is built on the data this wedge produces.
- **The moat is not the chatbot.** Free chatbots answer anything. Our moat is the structured paper bank, the official curriculum graph, the learner model, marking guides, local-language quality, and distribution through schools and mobile networks.
- **$3B ARR cannot come from Zambia.** Zambia is a proof market worth tens of millions of dollars at most. $3B is a ten-year, multi-country, multi-payer outcome (Section 7). It is useful as an architecture target, not as a forecast.
- **Build one loop, in one subject, and prove learning gains before adding anything.** Section 6 defines that loop. If it does not raise exam scores and people do not pay, no extra feature will save the company.

One-sentence product

> **A personal tutor, past-paper coach and study planner that works offline, explains in your own language, costs less than a loaf of bread a week, and proves to you, your parents and your school that you are improving.**

Tier	Meaning
MVP	Built in the first 90 days, one subject, 400-learner pilot
V1	Months 4 to 12: more subjects, channels and the habit layer
V2	Year 2: opportunity engine, marketplace, deeper offline
V3	Year 3 and beyond: new levels, countries, credentials

2. The Expensive, Headache-Giving Problems

These are the recurring problems learners (and the families paying for them) face. They are ordered by how much money, time or opportunity they cost. Cost descriptions are qualitative: they must be measured in the pilot (see the data register).

#	Problem	Real cost	Synapse answer	Tier
1	Private tuition is unaffordable	Extra lessons are the main route to better marks; families without money simply go without. Inequality compounds each year.	AI tutor with hint ladder at a fraction of the price; later a verified human-tutor marketplace paid by mobile money.	MVP
2	Past papers are scarce, costly and have no worked answers	Photocopied bundles, WhatsApp PDFs with no solutions or marking points. Learners practise blind.	Question-level paper bank with worked solutions and examiner points; tap any question for the solution.	MVP
3	Not knowing what you do not know	Time spent re-reading what is already known; surprise at results.	Short placement test, competence-level mastery map, one clear "do this next".	MVP
4	One-shot, high-stakes exams	A failed Grade 9 or 12 closes pathways and costs rewrite fees and a lost year.	Timed mock exams in real ECZ format, honest readiness score, rewriter track.	MVP
5	Language gap	Concepts taught in English only; many learners memorise without understanding.	Explanations and audio in Bemba and Nyanja first, other languages later; bilingual glossaries.	MVP
6	Data, power and shared phones	Video-based apps burn bundles; studying stops when signal or electricity does; one phone serves a household.	Offline packs, tiny downloads, audio, low-end Android, shared-device mode, WhatsApp and SMS lite channels.	MVP
7	Large classes mean no personal feedback	Questions are never asked; homework is rarely marked in detail.	Instant marking with explanations; ask anything privately without embarrassment.	MVP
8	Forgetting and last-night cramming	Knowledge fades; cramming only lasts to the exam.	Automatic spaced repetition and a mistakes notebook.	MVP
9	Wrong answers and untrustworthy AI	A confident wrong answer costs marks; free chatbots are not aligned to ECZ.	Curriculum-grounded answers with citations, maths verified by a symbolic engine, visible confidence and a report button.	MVP
10	Parents cannot help with homework	Children study alone; parents guess.	Family Link, tonight-conversation starters, one-line progress messages (Parent World).	V1
11	Distraction and procrastination	The phone competes with study; no routine.	Small daily goals, focus blocks, streaks with rest days, SMS and WhatsApp nudges.	V1
12	Missed school (fees, illness, work, pregnancy, distance)	Learners fall behind permanently and often drop out.	Catch-up pathway, absent-lesson recaps, return-to-learning plan.	V1
13	Exam stress and isolation	Anxiety, low confidence, especially for girls and rural learners.	Opt-in check-ins, exam-stress toolkit, peer groups, human escalation protocol.	V1
14	Out-of-school learners and rewriters	A large group with zero school support and the strongest willingness to pay.	Fully self-serve track with mocks and readiness; no class code required.	V1
15	Unclear future: subjects, colleges, bursaries	Wrong subject choices, missed deadlines, application fees wasted.	Career lens, subject-choice advisor, points calculator, bursary and deadline finder, application kit.	V2
16	No labs for practical subjects	Science, ICT and design taught as theory only.	Guided simulations and virtual practicals (a short list of excellent ones).	V2
17	Disability and accessibility	Materials do not fit learners with visual, hearing or reading difficulties.	Text-to-speech, large text, dyslexia-friendly font, captions, low-literacy icon mode.	V2
18	No proof of learning; lost certificates	Paper is lost; skills cannot be shown to colleges or employers.	Pathway Passport: a verified, shareable digital record.	V2

3. Broadening the Idea

The original idea is a study app. That is too narrow and too easy to copy. The broader idea: **the learner's operating system**. A single place that tells a learner where they stand, what to do today, helps them when stuck, proves they are ready, opens doors, and keeps them connected to people who help. Six layers, each answering a question the learner already asks.

Layer	Question the learner asks	What Synapse provides
STAND	Where am I really?	Placement test, mastery map, prerequisite gaps, readiness score
DO	What should I do today?	10, 20 or 30 minute daily plan from weak competences and exam date
UNDERSTAND	I am stuck.	Tutor with hint ladder, local-language and audio explanations, snap and solve
PROVE	Am I ready?	Timed mocks, auto-marking, mistakes notebook, spaced review
OPEN	What is next for me?	Career lens, subject advice, bursaries, applications, Pathway Passport
BELONG	Who is with me?	Study groups, verified tutors, family link, counsellor escalation

Broaden the delivery, not just the features

Most learners will not have data or a smartphone every day. A learning product that only lives in a full app reaches a small slice. So the same learner model is delivered through five channels:

Channel	Use	Tier
Android lite app	Full experience; small download; offline packs; works on low-end phones	MVP
WhatsApp bot	Daily question with explanation, "ask a question", mock reminders; cheap and familiar	V1
SMS and USSD	Reminders, weekly quiz, result checks, parent summaries; no smartphone needed	V2
Audio (revision radio)	Topic summaries in local languages, downloadable; works hands-free	V1
School Box	A small offline server in school: content, mocks and sync over local Wi-Fi	V2

Broaden the customer

- **Learner**: free core, pays small amounts for exam-season passes.
- **Parent or sponsor**: pays for a child, sees verified progress.
- **School**: pays for management, branding and analytics (School World).
- **Telco**: zero-rated data and carrier billing in exchange for a growing user base.
- **Government, donors, NGOs**: pay per learner for exam-prep programmes with evidence of gains.
- **Universities, colleges, employers, bursary funds**: pay for qualified, verified applicants.

4. Everything in the Learner's World

The complete inventory, grouped into fifteen pillars. Nothing from Master Plan v2 was dropped; items were merged, renamed where needed and given a tier. Items marked V2 or V3 are real but must not be built before the MVP loop is proven.

A. Home (Today)

Feature	What it does	Tier
Daily plan	Auto-built 10, 20 or 30 minute plan from weak competences, exam date and time available	MVP
Resume anywhere	Pick up exactly where you stopped, on any device, online or offline	MVP
Exam countdown and readiness	Days to each paper and an honest readiness score per subject	MVP
Streaks with rest days	Weekly goals that do not punish power or data outages	V1
Low-energy mode	Learner says they are tired; plan switches to lighter review tasks	V2

B. Diagnose (Where do I stand?)

Feature	What it does	Tier
Placement test	15-minute adaptive test per subject; finds true level including earlier-grade gaps	MVP
Mastery map	Competence-level green, amber, red with official codes (e.g. 4.1.4.1)	MVP
Prerequisite gap finder	Traces a failure back to the root concept it depends on	V1
Readiness estimate	Predicted grade band with a confidence range, shown honestly	V1

C. Tutor (Synapse AI)

Feature	What it does	Tier
Grounded chat with citations	Answers cite the official competence code and the paper or syllabus source	MVP
Hint ladder (Guide mode)	Hint, then bigger hint, then worked step; final answer only after real attempts	MVP
Explain and Practise modes	Short explanation, then immediate practice questions on the same competence	MVP
Verified-answer layer	Maths via symbolic engine, facts via the bank; otherwise the tutor says "I am not sure" and flags it	MVP
Bemba and Nyanja explanations	Ask in one language, revise in another; bilingual glossaries	MVP
Snap and Solve	Photo of a problem; solved by the engine; steps explained	V1
Voice tutor	Talk it through, hear the answer; hands-free for low-data homes	V1
Explain-it-back check	Learner explains in their own words; tutor checks understanding	V1
More languages	Tonga, Lozi, Kaonde, Lunda, Luvale, then regional languages	V1
Assess, Create, Organise, Plan modes	Remaining modes of the eight-verb engine	V2

D. Practice and Exams

Feature	What it does	Tier
ECZ paper bank	Real papers segmented question by question, tagged to competences, worked solutions, examiner points	MVP
Paper Mode	Open a paper; tap a question for solution, marking guide and a mini-quiz on that topic	MVP
Topic drills	Targeted practice on one competence with instant feedback	MVP
Personalised mocks	Timed, ECZ-format, from real bank questions with number variants; auto-marked	MVP
Mistakes notebook	Every wrong answer logged with error type and fixed by later review	MVP

Feature	What it does	Tier
Spaced repetition	Flashcards scheduled automatically (next day, three days, a week, and so on)	MVP
Full exam simulator	Strict timing, no hints, offline-capable	V1
Handwritten answer marking	Photo of working; OCR and marking against examiner points; teacher override	V1
Topic frequency map	How often each topic appeared over ten years, to focus revision	V1
Examiner tips	Where marks are commonly lost, by question type	V1

E. Library and Notes

Feature	What it does	Tier
Formula sheets and glossaries	Generated from the syllabus, per subject and grade	MVP
Notes, folders, PDF reader	Highlight and annotate; works offline	V1
Scan handwritten notes	OCR of photos into searchable notes	V1
Auto companions	Each note gets a summary, key points, flashcards and a mini-quiz	V1
Ask My Library	Answers grounded in the learner's own materials with page citations	V2

F. Study With Others

Feature	What it does	Tier
School-scoped study groups	Shared notes and decks, group quizzes, moderated threads	V1
Group challenges	Opt-in leaderboards that never shame low scorers	V1
Verified tutor marketplace	University students and teachers, identity-checked, paid via mobile money, sessions logged for safety	V2
Teach-to-learn credits	Learners earn credits by explaining topics to peers	V3

G. Planner, Focus and Habits

Feature	What it does	Tier
Revision plan generator	"ECZ Maths in 21 days" turned into a paced daily plan	MVP
Task planner	Goals and tasks, auto-filled with weak-topic practice and reviews	MVP
Focus blocks	Study timer with notification silencing using standard Android settings	V1
Habit nudges	SMS or WhatsApp reminders at the learner's chosen time	V1

H. Learn Anywhere

Feature	What it does	Tier
Offline topic packs	Small downloads: explanations, questions, solutions, audio	MVP
Low-data mode	Compressed images, no auto-playing media, data-use meter	MVP
Shared-phone mode	Several learners on one device, each with a PIN and separate progress	V1
Revision radio	Any topic as a short spoken summary in local languages	V1
WhatsApp bot	Daily question and explanation; ask-a-question	V1
USSD and SMS lite	Reminders, quizzes and result checks without a smartphone	V2
School Box	Offline local server for schools with poor connectivity	V2

I. Pathways and Opportunity

Feature	What it does	Tier
Career lens	Where each competence leads (nursing, engineering, accounting and more)	V1

Feature	What it does	Tier
Subject-choice advisor	Grade 9 to 10 and senior-secondary subject combinations, based on mastery and goals	V2
Points and requirements calculator	Shows which courses a predicted result unlocks	V2
Bursary and scholarship finder	Deadlines, eligibility, reminders	V2
Application kit	CV, personal statement coach, form guidance	V2
Pathway Passport	Verified, shareable digital record of mastery and certificates	V2
Skills tracks	Digital skills, entrepreneurship, financial literacy	V3

J. Wellbeing and Support

Feature	What it does	Tier
Mood and energy check-ins	Optional, learner-controlled; helps schedule hard topics	V1
Exam-stress toolkit	Breathing, planning, sleep and last-week routines	V1
Safe-reporting and escalation	Clear path to a human counsellor or trusted adult for bullying, abuse or crisis	V1

K. Inclusion and Access

Feature	What it does	Tier
Out-of-school and rewriter track	Full self-serve path with no class code	V1
Catch-up mode	After absence, a recap and short plan to get back on pace	V1
Accessibility suite	Text-to-speech, large text, dyslexia font, captions, low-literacy icon mode	V1

L. Rewards and Achievements

Feature	What it does	Tier
Badges and term certificates	School-branded where the school is on Synapse	V1
Opt-in leaderboards	Class level, personal-best framing	V1
Sponsored airtime or data rewards	Telco or sponsor funded for verified improvement	V2

M. Wallet and Payments

Feature	What it does	Tier
Free core always	Placement, daily plan, basic practice and offline packs stay free	MVP
Mobile-money passes	Day, week and exam-season passes via MTN, Airtel or Zamtel mobile money	MVP
Parent-pay and sponsor-pay	Family Link payments; sponsor-funded seats; voucher codes sold by agents	V1

N. Trust, Safety and Integrity

Feature	What it does	Tier
Minor-safe defaults	Guardian consent logic, content filters, no covert monitoring	MVP
Report-wrong-answer loop	One tap; human review; fixes feed the bank	MVP
Solution locks and audit	Teacher-assigned work locked until due; views logged	V1
Data controls	Learner and guardian can download or delete data	V1

O. Me and Family Link

Feature	What it does	Tier
Profile and language	Name, class, language, offline settings	MVP
Family Link	Guardian connects with the learner code; per-child visibility only	MVP
Notification controls	Quiet hours, channels, per-type settings	MVP

5. Fighting the Plan (Red Team)

Fifteen serious attacks on the Learner World plan, with the honest answer to each. If an answer is weak, that is where to spend effort first.

Attack	Why it stings	What we do about it
Ten spaces and sixty features means you will build nothing well.	True. Small teams die of breadth.	Ship one loop in one subject (Section 6). Everything else waits behind evidence.
Free ChatGPT, Gemini and Gauth already answer everything.	True for generic answers.	Compete where they cannot: official ECZ paper bank and marking points, learner model, local-language quality, offline packs, school and parent loop, mobile-money pricing.
Digitising syllabi and ECZ papers is a bottleneck and a copyright risk.	Reproducing exam papers may need permission from the Examinations Council.	Start permission talks with ECZ and CDC in week 1. Use cleared papers, write parallel questions, keep human review of every tag and solution.
AI costs will destroy margins at scale.	Chat at high-end model prices for millions is ruinous.	Cache answers per competence; pre-generate explanations; small models for routine work; premium models for marking and hard cases. Budget about 0.10 to 0.20 USD per active learner per month.
Offline-first and large AI models do not mix.	A cheap phone cannot run a large model.	Offline means pre-generated packs (explanations, questions, solutions, audio). Live AI when online; queue questions offline. Small on-device models later.
Learners and parents will not pay.	Incomes are low; free alternatives exist.	Free core; tiny exam-season passes; sponsor, school, telco and donor payers; test willingness to pay in the pilot before building billing depth.
Learners only study at exam time (seasonality).	Largely true.	Treat the exam season as the revenue peak and school terms as the retention base through teacher assignments. Multi-subject plans extend the season.
Teachers and schools adopt slowly.	Yes, procurement is slow.	Learner-first distribution (WhatsApp, ambassadors, out-of-school learners). Teacher tools save hours but are not required for the MVP.
Focus shield and parent monitoring will break store rules and trust.	Accessibility-service abuse is restricted; surveillance of minors is touchy.	Use standard silence and do-not-disturb settings. Learner owns the feature; parents see summaries only. Drop vision verification from early versions.
Snap and Solve is a cheating machine.	It can be.	Default to the hint ladder and step explanations; teacher can lock solutions on assigned work; log solution views.
The $3B ARR claim is fantasy.	For Zambia alone, yes.	Multi-country, multi-payer architecture from day one; staged gates; honest probabilities (Section 7).
Teacher yes or no mastery entries are noisy.	Yes, they are thin.	Primary signal is the learner's own practice data; teacher entries are one input; calibrate against mocks.
Artifact generator and 3D models are shiny objects.	Fun, not urgent.	Defer to V2. Then build ten excellent simulations for the hardest topics rather than infinite generation.
Protecting children's data is a legal and reputational minefield.	Learners are minors; breaches would be fatal.	Collect the minimum, separate identity from learning data, guardian consent, on-device processing, legal review of Zambian data protection law (see data register).
One wrong answer in exam season kills the brand.	Plausible.	Maths verified by engine; human-reviewed golden solutions for the paper bank; confidence shown; fast correction loop; never present an unverified answer as certain.

6. What Will Actually Work: The Minimum Lovable Loop

Pass ECZ Mathematics in 60 days. One subject, one exam, one daily loop, one price. Grade 12 Mathematics is the right first target: high stakes, large demand, and answers can be verified by a symbolic engine. Computer Studies and Commerce from the MVP table in Master Plan v2 follow once the loop works.

The daily loop (about 20 minutes)

Step	What the learner does	Why it works
1. Warm-up	Five adaptive questions from their weakest competences	Starts with what matters; feeds the mastery model
2. Learn	A two-minute explanation (text or audio, own language) only for what they missed	Short, targeted, cheap to download
3. Practise	Three real past-paper questions with the hint ladder	Exam-style practice with support, not answers
4. Review	Flashcards from today's mistakes, scheduled for later days	Beats forgetting
5. Proof	Readiness score moves; guardian receives a one-line message	Motivation for the learner; trust and payment from the parent

Weekly: a Saturday timed mini-mock. Monthly: a full-paper mock. Final fortnight: exam-simulator mode and a calm-down plan.

The pilot that proves it

- 400 learners: about 200 in two schools and about 200 out-of-school rewriters.
- Matched control group using the same placement test and mock at the start and end.
- Eight-week programme; payment test at two price points; WhatsApp and app both offered.

Gates (all must pass before building more)

Gate	Target
Learning gain	Mock-score gain at least 10 percentage points over control after 8 weeks (set a realistic target after the baseline)
Engagement	At least 40 percent of enrolled learners active weekly in week 8; day-30 retention at least 25 percent
Willingness to pay	At least 8 percent of active learners buy a pass at 1 to 2 USD per month equivalent
AI cost	At most 0.15 USD per active learner per month
Answer accuracy	Audited sample of tutor and solution answers wrong less than 2 percent
Trust	Parent rating of visibility better than any previous channel (survey)

Team shape for the first 90 days

- Product and curriculum lead (owns the loop and the pilot)
- Two engineers: Android and backend (Flutter, Supabase)
- One AI and data engineer (retrieval, verification, evaluation)
- Three part-time Zambian teacher-reviewers (tags, solutions, language checks)
- One field coordinator for pilot schools and rewriters

Kill rule. If after two pilot cycles learners do not improve or do not pay, stop adding features. Fix the loop, the content quality or the price. More pages will not rescue a loop that does not work.

7. Business Model and the $3B Arithmetic

Revenue lines

Stream	Who pays	Price shape
Learner Plus passes	Learner or parent via mobile money	About 0.5 USD per week-pass in exam season; about 2 to 4 USD per month; price varies by country
Family plan	Parent	About 4 to 6 USD per month for several children and the Parent Academy
School subscription	School	Per learner per term, tiered (School World)
Telco partnerships	Mobile network	Zero-rated data, carrier billing, revenue share
Government and donor contracts	Ministries, development agencies, NGOs	Per learner licence for exam-prep programmes, tied to evidence
Marketplace and credentials	Tutors, universities, employers, bursary funds	Take rate on tutoring (15 to 20 percent); fees for verified applicant leads and credentials

Reality check: Zambia alone

Zambia has on the order of 4 to 5 million learners in school (verify against Ministry statistics). If 10 percent paid about 20 USD per year, that is roughly 9 million USD. Adding schools and a donor-funded programme might double or triple it. **Zambia is a proof market worth tens of millions at the very most, not billions.**

What $3B ARR would take (one illustrative combination)

Revenue line	Volume assumption	Unit value	Annual
Consumer subscriptions	20 million paying learners	60 USD per year	1.2 B USD
School subscriptions	250,000 schools	1,000 USD per year	0.25 B USD
Government and donor programmes	100 million learners covered	9 USD per learner	0.9 B USD
Marketplace, credentials, sponsors, telco share	Multiple markets	Blended	0.65 B USD
Total			**3.0 B USD**

For scale: Duolingo, the leading consumer education company, reached roughly one billion dollars in annual revenue with on the order of ten million paying subscribers (figures from my last reliable information; check current numbers). The table above needs about twice that paying base plus large public-sector contracts, in markets with far lower incomes. That is why it is a ten-year, multi-country ambition and not a forecast.

What must be true

- **Evidence of learning gains** strong enough for governments and donors to buy (aim for trial-grade evidence).
- **A content engine** that can onboard a new curriculum, exam board and language in weeks. This ingestion pipeline is the real moat.
- **Distribution through telcos, schools and ministries**, not only app stores.
- **Serving cost under 1 USD per learner per year** on free tiers.
- **Quality AI in local languages**, measured and audited.

Milestone ladder (targets, not forecasts)

Stage	Markets	Paying learners	ARR target	Proof needed
Year 1	Zambia pilot	3,000	About 0.1 M USD	Learning gain, retention, payment test
Year 2	Zambia national	100,000	2 to 3 M USD	Repeatable acquisition below 2 USD per learner
Year 3	3 to 4 neighbouring countries	500,000	About 12 M USD	New country onboarded in under 3 months

Stage	Markets	Paying learners	ARR target	Proof needed
Year 5	8 to 10 African countries; first government contracts	3 million	About 120 M USD	Independent evaluation of gains
Year 7	Pan-Africa and first South or Southeast Asia market	10 million	About 600 M USD	Multi-payer model at scale
Year 10	Multi-region	20 million consumer plus public-sector programmes	Up to 3 B USD	Everything above holds

Honest probability. A 100 to 300 million USD ARR company across five to eight countries is a strong, plausible outcome if the loop works. $3B is the ambition that should shape architecture (multi-country, multi-language, multi-payer), but it should never justify building ahead of evidence.

Unit-economics targets

Metric	Target
Acquisition cost per learner	Under 2 USD through schools, WhatsApp referral and ambassadors
Gross margin	Over 70 percent
AI and hosting cost per active learner per month	At most 0.15 USD
Free to paid conversion of active learners	5 to 10 percent
Payback on acquisition	Under 3 months

8. Roadmap

When	What gets built or proven
Weeks 1 to 2	Choose the wedge subject; open permission talks with ECZ and CDC; obtain the syllabus and three years of papers; recruit teacher-reviewers; define pilot measurement (see data register)
Weeks 3 to 6	Digitise syllabus into the curriculum graph; ingest and tag papers; write and verify worked solutions; build placement test and mastery model; tutor with hint ladder and verified maths
Weeks 7 to 10	Android lite app; offline packs; mocks and mistakes notebook; spaced repetition; mobile-money pass; WhatsApp daily question
Weeks 11 to 13	Run the 400-learner pilot; collect baseline and endline; audit answers; test two price points
Months 4 to 6	Second and third subjects; Family Link and parent messages; rewriter marketing; Bemba and Nyanja audio
Months 7 to 12	Focus blocks, study groups, handwritten marking, voice tutor, shared-phone mode, first school contracts, sponsor seats
Year 2	Opportunity engine (subject advisor, bursaries, applications), tutor marketplace, USSD, School Box, Pathway Passport, second country preparation
Year 3+	Primary and tertiary content, skills tracks, new countries, government programmes, credentials

9. Metrics

Type	Metric
North Star	Verified mastery points gained per learner per week
Outcome	Mock and ECZ result improvement against control; competences moved from red to green
Engagement	Weekly active learners; day-7 and day-30 retention; loop completion rate
Revenue	Paying learners; conversion; ARPU; refund and failed-payment rate
Quality	Wrong-answer rate on audited samples; reports per 1,000 answers; fix time
Reach	Share of learners on offline or WhatsApp channels; data used per learner
Trust	Guardian link rate; parent satisfaction; consent withdrawal rate

10. Do These This Week

- Pick the wedge: ECZ Grade 12 Mathematics (or a different single subject with clear reasons).
- Write to ECZ and CDC asking for permission or a partnership to use past papers and the syllabus.
- Get one syllabus PDF and three years of papers for that subject and run the first ingestion.
- Recruit three teacher-reviewers and one rewriter group to interview.
- Draft the pilot protocol and the baseline test; use the data register for exactly what to log.